The ~1830 Ma Trans-Hudson hairpin from paleomagnetism of the Wapisu gneiss dome, Kisseynew Domain, Manitoba[1]

D.T.A. Symons and M.J. Harris

Abstract: The Wapisu gneiss dome is located in the northeastern part of the Kisseynew Domain of the Trans-Hudson Orogen (THO) in north-central Manitoba. The dome is circular, about 6 km in diameter, with steeply-dipping flanks. It is composed of upper amphibolite-facies gneisses derived from turbiditic sediments, with leucogranitic sill-like intrusions that were metamorphosed, starting at about 1830 Ma. Alternating field and thermal step demagnetization of 153 specimens from 17 sites around the perimeter of the dome isolated a characteristic remanent magnetization (ChRM) direction of D = 346.6°, I = 78.9° (α_{95} = 5.3°, k = 46, N = 17) that gives a pole of 119.5° W, 75.8° N (d_p = 9.5°, d_m = 10.0°). Unblocking temperatures and saturation isothermal remanence analyses show that the ChRM resides mostly in single to pseudosingle domain magnetite or titanomagnetite, with minor hematite commonly present and with minor pyrrhotite present in the leucogranites. The ChRM is found to be postfolding with >>99.9% confidence, indicating acquisition on cooling from peak metamorphism at ~1810 ± 10 Ma. This Wapisu gneiss dome paleopole is the first from the Kisseynew Domain and the first from the 1830 to 1770 Ma interval in the THO. It indicates an ~90° bend or hairpin in the apparent polar wander path for the THO juvenile terranes and Superior craton. It is speculated that the hairpin marks the collisional impact of the Archean Sask craton and (or) Hearne craton, which drove the Paleoproterozoic Kisseynew Domain into the Archean Superior craton.

Résumé : Le dôme de gneiss de Wapisu est localisé dans la partie nord-est du Domaine de Kisseynew de l'orogène trans-husonien au centre-nord du Manitoba. Le dôme est circulaire, son diamètre est d'environ 6 km et ses flancs sont fortement inclinés. Il est composé de gneiss du faciès des amphibolites supérieur dérivés de sédiments turbiditiques, incluant des intrusions leucogranitiques ressemblant à des filons-couches qui furent métamorphisés; le métamorphique ayant débuté vers 1830 Ma. La désaimantation par champ alternatif et chauffage par paliers successifs de 153 spécimens, provenant de 17 sites sur le pourtour du dôme, a permis d'isoler une aimantation rémanente caractéristique avec direction D = 346,6°, I = 78,9° (α_{95} = 5,3°, k = 46, N = 17), fournissant un pôle de 119,5° W, 75,8° N (d_p = 9,5°, d_m = 10,0°). Les températures de déblocage et les analyses de la saturation rémanente isothermique montrent que l'aimantation rémanente caractéristique réside principalement dans les domaines monocristallins ou pseudo-monocristallins de la magnétite ou la titanomagnétite, avec habituellement une faible proportion d'hématite et une faible quantité de pyrrhotite dans les leucogranites. L'acquisition de l'aimantation rémanente caractéristique est estimée postérieure à la déformation avec un niveau de confiance >>99,9%; cette acquisition daterait de la phase de refroidissement qui a succéda l'apogée métamorphique, il y a ~1810 ± 10 Ma. Ce paléopôle du dôme de gneiss de Wapisu est considéré comme étant le premier du Domaine de Kisseynew et le premier de l'intervalle 1830 à 1770 Ma dans l'orogène trans-hudsonien. Il indique un tournant, ou un virage en épingle à cheveux, dans la courbe de dérive des pôles apparente associée aux terranes juvéniles de l'orogène trans-hudsonien et au craton du Supérieur. On suppose que le virage en épingle à cheveux marque l'impact de la collision du craton Sask et (ou) craton Hearne de l'Archéen, qui a poussa le Domaine de Kisseynew paléoprotérozoïque dans le craton du Supérieur archéen.

[Traduit par la Rédaction]

Introduction

The Kisseynew Domain underlies a larger area in northern Manitoba than any other domain in the Trans-Hudson Orogen (THO) (Fig. 1). As such, its location during the latter stages of collision of the Archean Hearne craton with the Archean Superior craton in the Paleoproterozoic is critical for understanding the geotectonic evolution of the intervening THO. However, it is not an easy task to decipher its relative motion, because the Kisseynew Domain is composed substantially of moderate to high-grade metamorphic rocks that are apparently devoid of significant ore deposits. One consequence is that most geologic mapping in the domain is at a reconnaissance or regional scale, and a second consequence is that rock units in the domain have not been a target for previous paleomagnetic studies.

Traditionally, paleomagnetists have avoided metamorphic rocks, because they often give uninterpretable results. High-

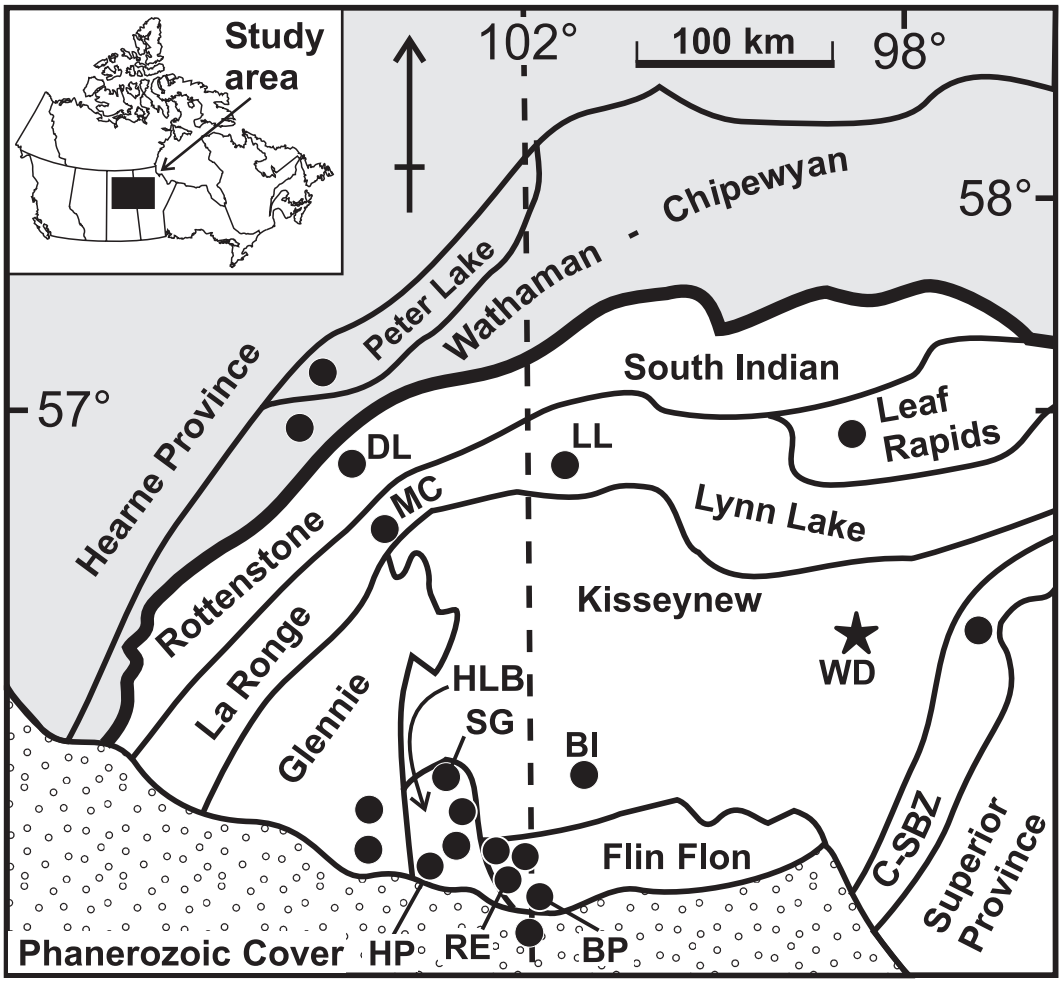
Received September 25, 1998. Accepted April 12, 1999.

D.T.A. Symons.[2] Earth Sciences, University of Windsor, Windsor, ON N9B 3P4, Canada.
M.J. Harris. Department of Earth Sciences, University of Western Ontario, London, ON N64 5B8, Canada.

[1]Lithoprobe Contribution 1157.
[2]Corresponding author (e-mail: dsymons@uwindsor.ca).

© 2000 NRC Canada

Fig. 1. Tectonic elements of the Trans-Hudson Orogen and location of the Wapisu gneiss dome. C-SBZ, Churchill–Superior boundary zone; HLB, Hanson Lake block. Dots and star show locations of paleomagnetic collections, with those referred to in the text labelled as follows: BI, Big Island gneiss dome; BP, Boot–Phantom granites; DL, Davin Lake pluton; HP, Hanson Lake pluton; LL, Lynn Lake gabbro pipes; MC, Macoun Lake pluton; RE, Reynard Lake pluton; SG, Sahli granulite; WD, Wapisu gneiss dome.



grade granulitic rocks are usually an exception to this generalization and have given useful results both in the THO (Gala et al. 1998; Halls 1998) and elsewhere in Precambrian rocks (Costanzo-Alvarez and Dunlop 1988; Symons and Vandall 1990). This is because granulites have been entirely recrystallized and heated above the unblocking temperature spectra of the constituent magnetic minerals, and thus their characteristic remanent magnetization (ChRM) commonly records cooling from peak metamorphism through the 675° to 500°C blocking temperature range in hematite and magnetite during orogenic uplift and exhumation. Low to medium-grade metamorphic rocks are more problematic, because they tend to superimpose secondary magnetic components over the original primary remanence as new magnetic minerals are formed or as existing magnetic domains are realigned during metamorphic heating and retrograde cooling.

The other major problem for paleomagnetists in metamorphic rocks is the question of the paleohorizontal (i.e., how do you know that the study area has not undergone a regional tilt since the ChRM was acquired that would result in an erroneous pole position?). One approach is to sample over a large area so that localized tilts are averaged. A second approach is to use geothermobarometry, if a suitable mineralogic assemblage is present (e.g., Ague and Brandon 1992; Harris et al. 1997). The approach used in this study was to look for a nearly symmetrical gneiss dome or basin in which a well-defined lithologic unit has been observed to dip uniformly away from the crest or towards the trough. The implication is that the unit was originally stratiform and has subsequently been symmetrically folded. This approach has been previously used in the THO, both successfully (e.g., Gala et al. 1998) and unsuccessfully (Symons and Harris 1998), and is successful for the Wapisu gneiss dome. Despite the risks of failure, it is important to point out that many major geotectonic problems will remain unsolved as long as paleomagnetists continue to avoid metamorphic terranes rather than learning how to extract information from them.

Lithoprobe has sponsored a series of paleomagnetic studies over the past few years in the THO, mostly in the laboratory at University of Windsor. The results to date have recently been summarized by Symons (1998) and Symons and MacKay (1999). For this study, there is one salient result: apparent polar wander path (APWP) segments in the

© 2000 NRC Canada

~1890 Ma to ~1830 Ma time span for the Superior craton, Flin Flon Belt, Hanson Lake block and Lynn Lake – LaRonge Domain all project to convergence at ~1830 Ma. This is the classical paleomagnetic signature for continental accretion tectonics, and these domains envelop the Kisseynew Domain on all sides (Fig. 1). Peak metamorphism in the Kisseynew Domain is shown by U–Pb geochronology to be in the 1820–1805 Ma interval (Gordon et al. 1990; Ashton et al. 1991; David et al. 1996; Ansdell and Norman 1995; Ansdell et al. 1995). Thus the ChRM for the Wapisu gneiss dome, which is both coherent and postfolding, should record acquisition at ~1810 ± 10 Ma and locate the assembled terranes at that time.

Geology

Zwanzig (1990) and Ansdell et al. (1995) have recently provided geologic overviews for the Kisseynew Domain. It comprises 1855 to 1841 Ma or somewhat younger turbidites and continental sandstones (Machado and Zwanzig 1995), with rare volcanic horizons that were intruded by 1840 to 1830 Ma plutons. During and after plutonism these rocks were subjected to dominantly high-temperature and low-pressure metamorphism that reached moderate to high metamorphic grades and peaked between 1820 and 1805 Ma. Associated with metamorphism was extensive ductile fold-and-thrust deformation that collapsed the domain to a fraction of its original areal extent, as part of terminal continental collision (Lucas et al. 1994), often placing higher grade rocks over lower grade rocks to form a metasedimentary gneiss belt.

The Wapisu Lake area was mapped geologically at a scale of 1 : 50 000 by T.G. Frohlinger in 1971 and 1973, as part of the Burntwood Project (Map 78-3-13, Wapisu Lake, in Baldwin et al. 1979). The description of the rock units and their nomenclature follows that used for the recent Bedrock Compilation Map Series (Manitoba Energy and Mines 1989). On both maps the gneiss dome appears as an oval structure of about 6 km in diameter on the northern flank of the Kisseynew Domain (Fig. 2). The dome is composed of rocks of the Burntwood River metamorphic suite that are divided into two units. One unit (labeled "Gneiss, from mudstones" in Fig. 2) is a graphitic biotite–quartz–plagioclase gneiss, with or without cordierite or sillimanite, that was derived from greywacke and mudstone. The other unit (the "Migmatite, from greywacke") is composed of migmatite, with up to 75% anatectic mobilizate, that was derived from greywacke. Where sampled, the mobilizate is aplitic to pegmatitic segregations of garnetiferous tonalite to granite. The "Gneiss, from sandstone" unit (in Fig. 2) is thought to represent the Sickle metamorphic suite. It is a magnetiferous quartzofeldspathic gneiss, locally including cordierite, sillimanite, or garnet, that was originally a sandstone. Three phases of Paleoproterozoic intrusive rocks are recognized that cut the metasedimentary gneisses. The most abundant is the white leucogranite with isolated garnet clots; this was sampled. The other two phases are grey hornblende–biotite tonalite to granodiorite and pink porphyroblastic granite with hornblende and magnetite.

Experimental methods

Sampling

The Wapisu gneiss dome was sampled at 17 shoreline sites around three-quarters of the perimeter of the dome (Fig. 2). Of the 17, 12 sites were in the more mafic "Migmatite, from greywacke" unit of the Burntwood River gneisses. The structure as mapped appears to be a dome or doubly plunging antiform with limbs dipping steeply away in all directions (Baldwin et al. 1979). The remaining five sites were from the later felsic "White Leucogranite" unit which appears to have been originally emplaced as a sill prior to doming.

At each site, five cores were drilled with a diameter of 2.54 cm and oriented in situ with a sun or magnetic compass. A total of 153 specimens, each with a length of 2.20 cm, were sliced from the cores (Table 1). The specimens were stored for three months in a steel-shielded room with an ambient field of ~0.2% of the Earth's magnetic field intensity to allow their viscous remanent magnetization (VRM) components to decay. All subsequent measurements were done without removing the specimens from the shielded room.

Natural remanence

First the natural remanent magnetization (NRM) of each specimen was measured on an automated Canadian Thin Films DRM-420 cryogenic magnetometer with a lower resolution limit of ~3 x 10^{-6} A/m. The migmatite specimens have relatively weak, but fairly uniform, NRM intensities of M = 2.0 (Q_1 = 1.3; Q_3 = 8.4) x 10^{-4} A/m where M = median; Q_1 = lower quartile; Q_3 = upper quartile. The leucogranite specimens are somewhat more intense with M = 4.2 (Q_1 = 1.3; Q_3 = 8.4) x 10^{-4} A/m.

Thermal step demagnetization

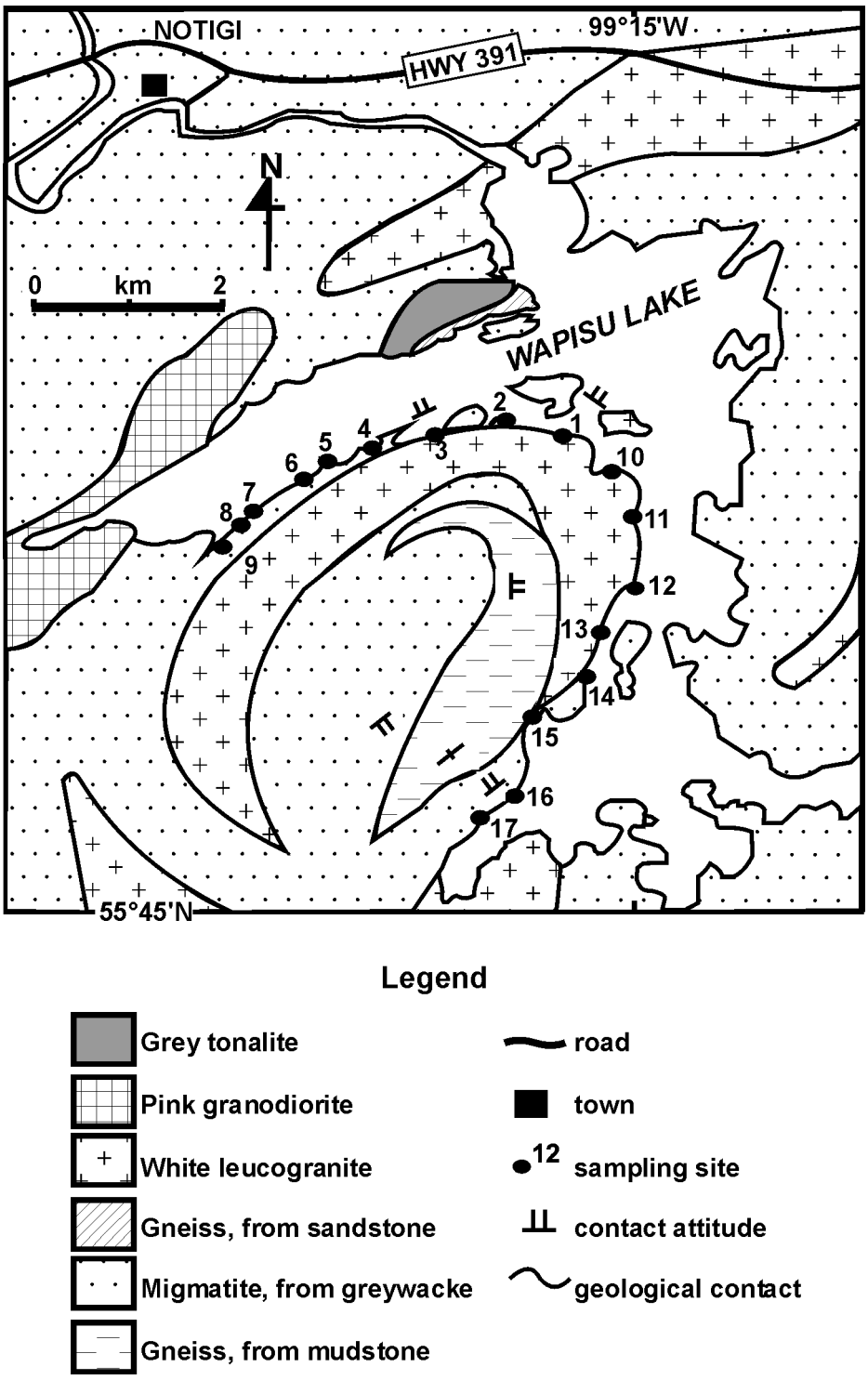
One pilot specimen per site was thermally demagnetized in 13 steps up to 585°C using a Magnetic Minerals MMTD-1 oven. Most of the specimens show distributed unblocking temperature spectra with preferential unblocking in the 450° to 585°C range that is indicative of magnetite or titanomagnetite (Fig. 3). The leucogranite specimens also show preferential unblocking between 250° and 325°C that indicates the presence of some pyrrhotite (Fig. 3*a*). After the initial removal of VRM by ~250°C, nearly all of the specimens isolate a ChRM direction that is inclined steeply downwards towards the north. Complete demagnetization is rarely achieved by 585°C, indicating that residual hematite is also present (Fig. 3*b*, *c*, *d*). However, the directions become erratic and intensities commonly increase after the 555°C to 570°C step, because new magnetic minerals are being formed in the oven by oxidation, preventing isolation of the hematite component.

Alternating field step demagnetization

A second pilot specimen per site was demagnetized in an alternating field (AF) in 12 steps up to 100 milliTesla (mT) using a Sapphire Instruments SI-4 demagnetizer. VRM was typically removed by the 20 mT step and a steeply-downward ChRM component was isolated, with a northerly

© 2000 NRC Canada

Fig. 2. Geology of the Wapisu gneiss dome simplified from Baldwin et al. (1979), with sampling sites.



bias (Fig. 4*a*, *b*) similar to the thermal results. A few specimens had a less stable remanence and demagnetized more quickly (Fig. 4*c*), and a few were very stable and resistant to AF demagnetization (Fig. 4*d*). The remainder of the specimens were AF demagnetized in six or more steps beginning at either 15 or 20 mT and ending at between 50 and 110 mT. Additional thermal steps were done in the 290° to 325°C or 500° to 550°C range in order to fully demagnetize some specimens.

Statistical analysis

The ChRM directions for individual specimens were obtained by visual inspection of orthogonal vector plots (As and Zijderveld 1958) and by least-squares analysis of the di-

© 2000 NRC Canada

Table 1. Site mean remanence data.

	Specimens			Coercivity		Temperature		Mean direction				
Site	m	n	r	Min. (mT)	Max. (mT)	Min. (°C)	Max. (°C)	*D* (°)	*I* (°)	α_{95}	*k*	*q*
1	9	9		20	100	200	520	331.6	77.2	4.6	125	+
2	8	4		20	55			312.8	76.7	16.2	18	
3	4	3		20	50	200	325	343.4	63.6	29.1	19	*
4	12	10		20	80	200	325	294.6	83.7	14.9	12	
5	11	11		20	80	200	540	5.9	83.7	6.9	44	+
6	13	13		20	100	200	555	288.1	82.3	6.8	39	+
7	10	9	1	20	50	275	325	285.4	76.3	34.1	3	*
8	9	8	1	20	60	325	525	228.6	81.4	11.4	21	+
9	5	3		20	50	200	520	34.6	75.6	15.9	61	
10	11	5	2	20	50			8.7	74.1	14.2	19	
11	6	6		20	70			342.8	56.0	9.1	55	+
12	6	5		20	55			338.7	81.2	16.1	24	
13	10	4		20	80			25.4	65.2	17.4	29	
14	13	7		20	50	250	300	11.6	77.4	15.5	16	
15	8	7		20	110			335.4	73.2	12.6	24	
16	10	6		20	100	200	555	345.6	81.2	15.3	20	
17	8	6		20	60	275	325	80.4	79.2	25.7	8	*
Mean A	17							346.6	78.9	5.3	46	
Mean B	5							328.8	78.2	14.4	29	
Mean C	14							347.5	78.8	5.5	53	

Note: Specimens, number of measured (m) and useable normal (n) and reversed (r) end points; Coercivity, minimum (Min.) and maximum (Max.) coercivities in milliTesla (mT) represented in mean; Temperature minimum (Min.) and maximum (Max.) unblocking temperatures in degrees Celsius (°C); *D*, declination; *I*, inclination; α_{95}, radius of cone of 95% confidence in degrees and and its precision parameter (*k*) (Fisher 1953) and quality (*q*) of clustering. Mean: A, all sites; B, best five sites (+); C, excluding worst three sites (*) in terms of quality.

rection vectors (Kirschvink 1980) to define the straight line segments. For the latter method the maximum angular deviation had to be less than 15° for acceptance of an endpoint, with most being much less than 5°. All of the end points had normal polarity, except for four that were reversed. It is probable that several specimens were rejected because they failed to reach stable reversed end points. The site mean ChRM directions were then calculated using Fisher (1953) statistics (Table 1). They form a coherent cluster directed steeply downwards to the north (Fig. 5).

Saturation remanence

To better characterize the magnetic minerals carrying the ChRM, seven representative specimens were pulse magnetized in a direct field up to 900 mT in 10 steps using a Sapphire Instruments SI-6 pulse magnetizer, and then AF demagnetized in eight steps to 100 mT. Six of the seven specimens showed rapid acquisition of remanence intensity, reaching >85% of their saturation isothermal remanent magnetization (SIRM) intensity by 350 mT (Fig. 6). This behaviour is typical of single domain to pseudosingle domain magnetite, titanomagnetite, and pyrrhotite. The slower climbs from 85% to 100% indicate the presence of some hematite. One specimen shows only the presence of hematite, because it climbs in the hematite field. The decay of the SIRM on AF step demagnetization also indicates that the domains are single to pseudosingle in size and the slight flattening of the curves from 70 to 100 mT again reflects a hematite contribution.

Fold test

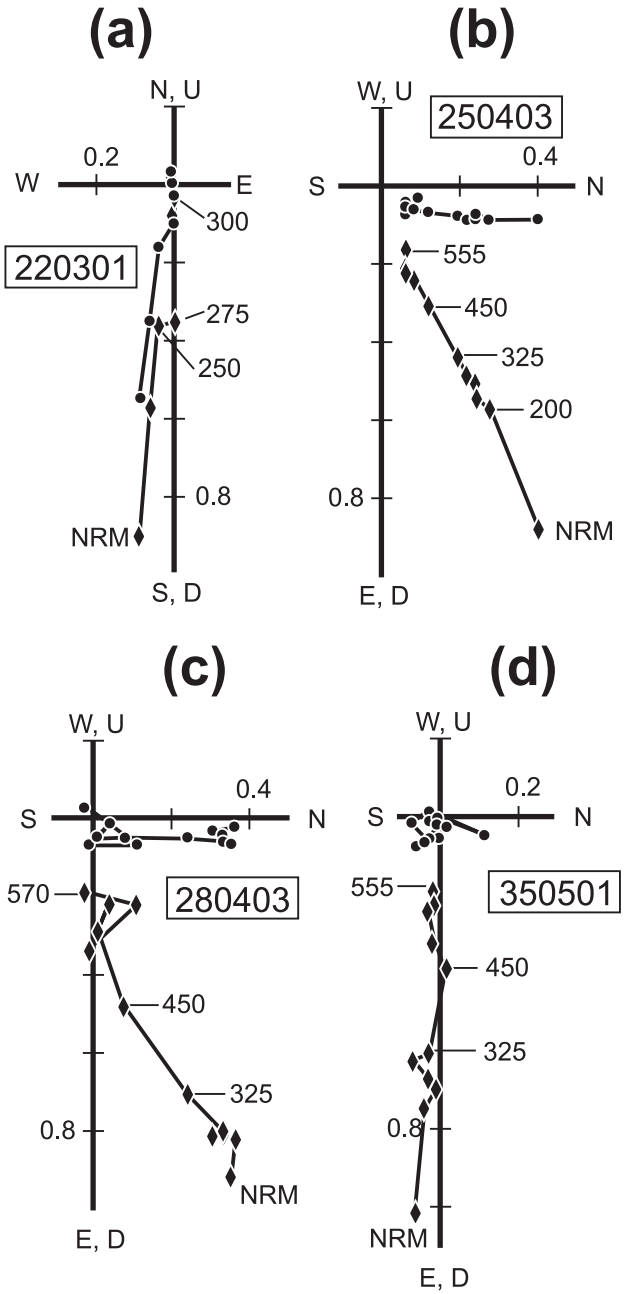
As the Wapisu domal structure is unfolded, the ChRM directions scatter, migrating to the perimeter of a stereonet with the precision parameter (*k*) of Fisher (1953) decreasing progressively from 46 to 1.4. Thus the paleomagnetic fold test (Graham 1949) shows that the remanence is entirely postfolding in origin, coincident with cooling from garnet-grade amphibolite facies metamorphism. Statistically, using the fold test of McElhinny (1964), which treats each site as an independent limb, the ratio of the precision parameter before and after unfolding (k_r) is 46/1.4 = 32.8, which greatly exceeds the comparison statistic (k_c) for 99.0% confidence of 2.20, showing that the ChRM is certainly postfolding in origin.

Unit mean direction and pole

Of the 17 sites, the specimen ChRM directions from three sites (3, 7, and 17) were poorly clustered with radii for their cones of 95% confidence (α_{95}) of >17.4°. Note, however, that their exclusion does not change significantly the mean direction or degree of clustering of the remaining 14 sites (Mean A versus Mean C, Table 1). Therefore the accepted mean direction is declination (*D*) = 346.6°; inclination (*I*) = 78.9°; α_{95} = 5.3°, *k* = 46, and the number of sites (*N*) = 17 (Fisher 1953). Using a mean site location of 99.28° W, 55.8°

© 2000 NRC Canada

Fig. 3. Orthogonal vector plots showing example specimens on thermal step demagnetization as described in the text. Axes are north (**N**), east (**E**), south (**S**), west (**W**), up (**U**) and Down (**D**). Axial values are expressed as a ratio of the specimen's natural remanent magnetization (NRM) intensity. Projections in the horizontal and vertical planes are circles and diamonds, respectively. Some step temperatures are labelled in degrees Celsius.

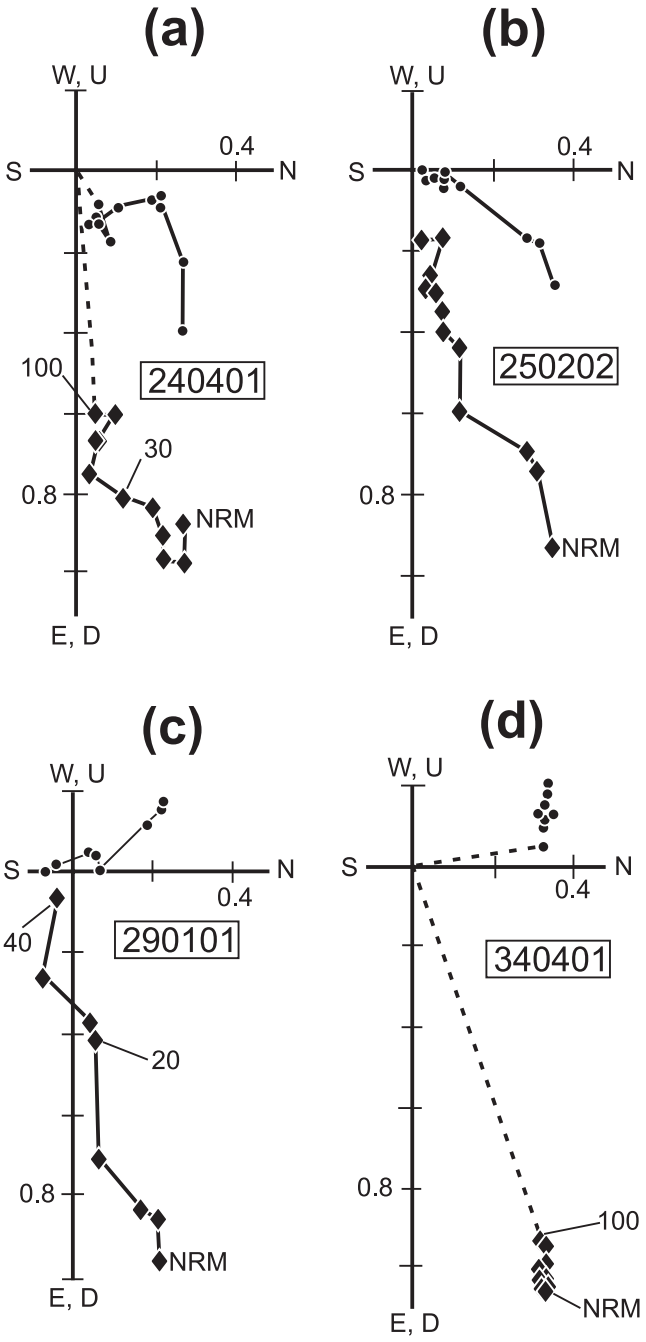


N, the calculated paleopole position is 119.5° W, 75.8° N (d_p = 9.5°, d_m = 10.0°) (longitude, latitude (radii of oval of 95% confidence)).

Discussion

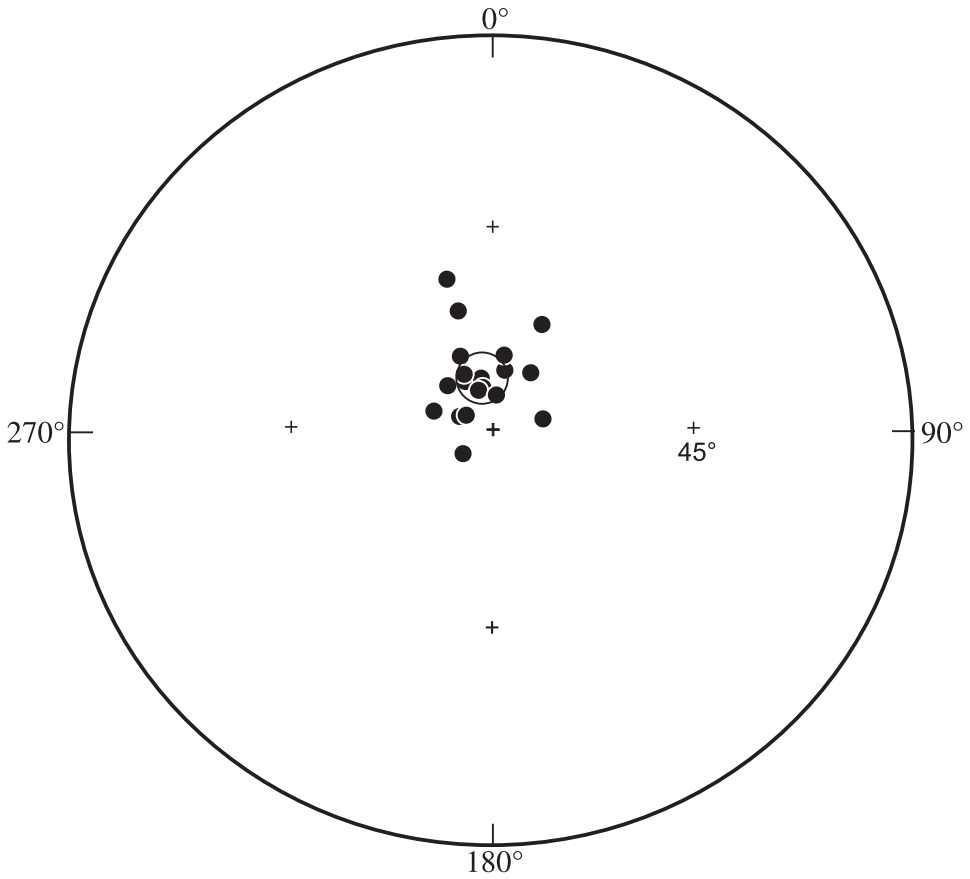
Relatively little radiometric age dating has been done on the rocks of the Kisseynew Domain and none has been done in the immediate area of the Wapisu gneiss dome. Nevertheless, the overall post-1830 Ma pattern for the domain seems clear: (1) U–Pb zircon dates for intrusive rocks that predate metamorphism are 1830^{+11}_{-5}, 1827^{+3}_{-7}, and 1826^{+11}_{-5} Ma; (2) U–Pb zircon dates for migmatites and mobilizates that record peak metamorphism are 1816^{+23}_{-12} and 1814^{+17}_{-11} Ma; (3) U–Pb monazite and titanite dates of 1817 ± 5, 1808 ± 2, 1806 ± 2, 1806 ± 2, 1804 ± 3, and 1804 ± 2 Ma record rapid postpeak metamorphic cooling; and

Fig. 4. Orthogonal vector plots of example specimens on alternating field step demagnetization as described in the text. Conventions as in Fig. 3, except that step fields are labelled in milliTesla.



© 2000 NRC Canada

Fig. 5. Equal-area stereogram showing the site mean remanence directions uncorrected for tilt, with the cone of 95% confidence for the unit mean.



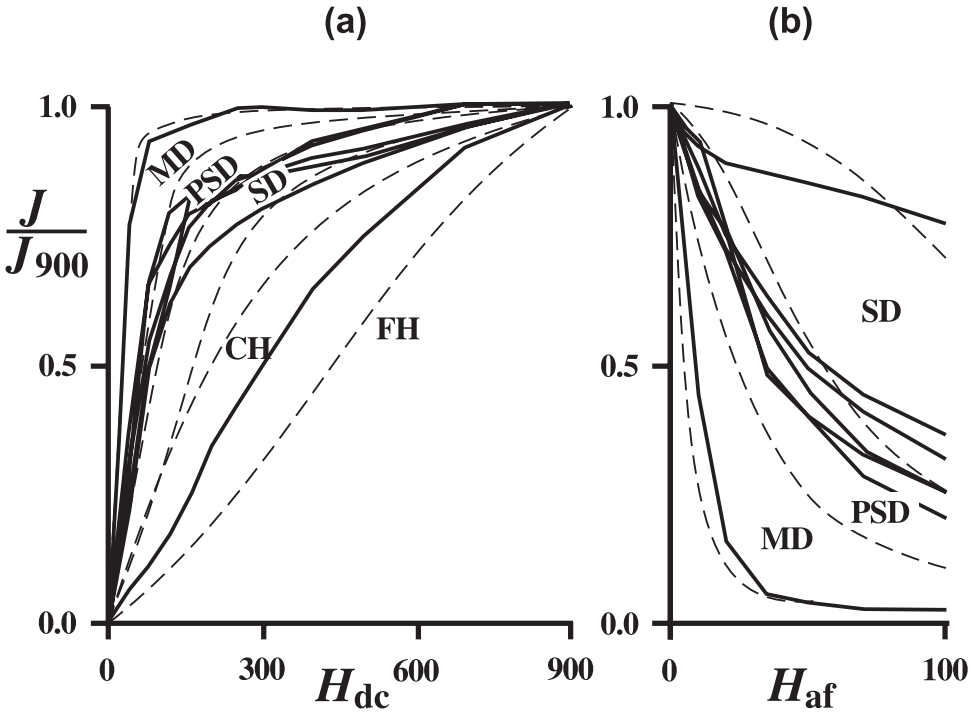
(4) U–Pb monazite dates for late pegmatitic granites and granitic pegmatites that record the terminal stages of metamorphism are 1801 ± 3, 1799 ± 3, and 1799 ± 2 Ma (Hunt and Roddick 1988; Gordon et al. 1990; Ashton et al. 1991; Hunt and Schledewitz 1992; Ansdell and Norman 1993). From this spectrum of ages and the blocking temperatures of the ChRM, it is probable that the ChRM of the Wapisu dome postdates 1815 ± 5 Ma and predates 1805 ± 5 Ma. Hence the assigned age for the ChRM is 1810 ± 10 Ma.

The Paleoproterozoic Manikewan ocean (Stauffer 1984) was about 5500 ± 700 km wide at ~1855 Ma (Dunsmore and Symons 1990; Symons 1998). Most of this width was between the 1856 Ma Wathaman batholith (Symons 1991) and the Rottenstone Domain (Fig. 1) (Symons et al. 1996). The batholith is deemed to be an Andean-type calc-alkaline continental-margin pluton (Meyer et al. 1992) that was emplaced along the margin of the Archean Hearne craton and pericratonic Peter Lake Domain on the northwest side of the southeast-verging Manikewan suture, because all of these terranes give subtropical paleolatitudes (Symons 1998). The terranes to the southeast of the suture in the Reindeer zone and the bounding Archean Superior craton all have given polar paleolatitudes at ~1855 Ma, indicating that these terranes were formed near to, and then assembled against, the craton (Symons and MacKay 1999). To date, rudimentary APWP segments are available in the 1890 to 1830 Ma time span for the (1) Superior craton from the 1877 Ma Molson dikes' C pole (Zhai et al. 1994; Halls and Heaman 2000) and 1849 Ma Sudbury irruptive complex pole (Morris 1984); (2) Flin Flon belt from the 1851 Ma Reynard Lake pluton (Symons 1995) and 1838 Ma Boot–Phantom pluton (Symons and MacKay 1999); (3) Hanson Lake block from the 1844 Ma Hanson Lake pluton (Gala et al. 1994) and the ~1830 Ma Sahli charnockite, derived from granulite facies metamorphism of the Archean Sask craton (Gala et al. 1998); and, (4) Lynn Lake – LaRonge belt from the vertical 1890 Ma Lynn Lake gabbro pipes (Dunsmore and Symons 1990) and the 1849 Ma Macoun Lake pluton (Symons et al. 1994) (Fig. 7*a*). These APWP segments project to a common pole at ~1830 Ma, thus defining the classical paleomagnetic signature or pattern for terranes accreting by "continent–continent" collision (Fig. 7*b*). Details of the sequential assembly of the THO terranes between the Archean craton during the ~1840 to ~1830 Ma interval are given by Symons and McKay (1999).

The pattern has obvious limitations. For example, the APWP segments are based on only two poles each and the Superior craton path in particular requires substantial extrapolation to reach the common pole. These limitations and the tectonic model that the segments predict are set out in Symons and MacKay (1999). The limitations of the Superior craton path in particular have led to further efforts, now in progress, to better define its path from ~1840 to ~1770 Ma (Harris et al. 1998). The pattern also elicits the obvious question — what happened from ~1830 Ma to 1770 Ma as

© 2000 NRC Canada

Fig. 6. Isothermal remanent magnetization (IRM) curves for example specimens showing (*a*) acquisition of IRM in a direct field (H_{dc}) in milliTesla; and, (*b*) demagnetization of IRM in an alternating field (H_{af}) in milliTesla. The intensity (J) is expressed as a ratio of the intensity at 900 mT (J_{900}). Shown are typical curves for single, pseudosingle and multidomain magnetite (SD, PSD, MD) and for fine and coarse grained hematite (FH, CH).



Trans-Hudson orogenesis came to an end? Two studies were undertaken in the Kisseynew Domain that focus on this question. The first, the Big Island gneiss dome on the southern flank of the Kisseynew Domain, proved unsuccessful as briefly reported by Symons and Harris (1998). This study represents a second attempt. Two additional studies are in progress on ~1800 Ma granites and on ~1770 Ma pegmatites in the Deschambault Lake area of the Glennie Domain.

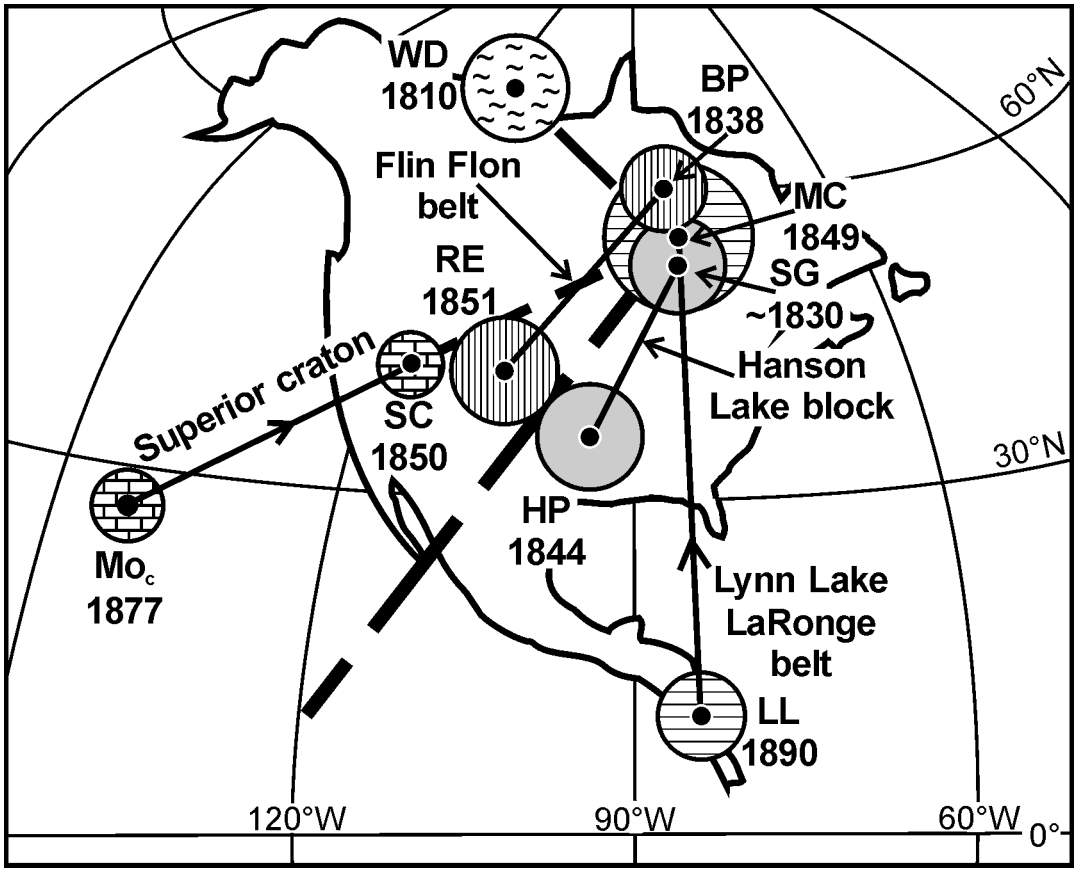
The pole position for the Wapisu gneiss dome is shown on Fig. 7. The fact that it is close to the present Earth's magnetic field direction is noted, but not considered a concern given the high levels of AF and thermal step demagnetization employed, given the evidence that single and pseudosingle domain magnetite and pyrrhotite carry the remanence, and given that the rocks are fresh from recent glaciation. At present, the pole position invites speculation rather than mandating firm conclusions. Specifically, it defines a sharp ~90° change in trend of apparent polar wander from northeast in today's coordinates between ~1890 Ma and ~1830 Ma to northwest between ~1830 Ma to 1810 Ma. Irving and Park (1972) noted that orogenic collisions inherently result in a dramatic change in a plate's motion, resulting in a sharp bend in the APWP that they termed a "hairpin." In the North American APWP, for example, the Laramide orogeny beginning at ~130 Ma, the Alleghanian orogeny at ~325 Ma, and the Acadian orogeny at ~375 Ma are all marked by ~90° hairpins (Besse and Courtillot 1991; Van der Voo 1993). If the existence of the hairpin is substantiated by further work, it will add credence to the notion that the existing outcrop extents of the various domains and belts in the THO are minor folded and thrust-fault telescoped remnants (Ansdell et al. 1995; Zwanzig 1997) of very much larger domains and belts, sufficiently large to cause the very-large Superior craton to change drift direction. This result suggests, therefore, that either the Hearne craton or the Sask craton or both cratons closed against the Kisseynew Domain, driving it into the Superior craton to complete accretion at this time.

Conclusions

It is concluded that a stable postdeformational ChRM has been isolated in the rocks of the Wapisu gneiss dome, and that it was acquired at 1810 ± 10 Ma ago as a thermoremanent remagnetization during cooling from upper amphibolite facies metamorphism. The presence of steep dips away from the crest of the dome in all directions of a more mafic layer of apparent stratiform origin is accepted as evidence that postmetamorphic tilting of the domal structure during cooling was negligible. The resulting pole position for the dome at 119.5° W, 75.8° N (d_p = 9.5°, d_m = 10.0°) is the first reported pole both for the Kisseynew Domain and for the ~1830 to ~1770 Ma period that spans from the peak to the denouement of the Paleoproterozoic Trans-Hudson orogeny. It is speculated that the pole may be the first recording of the postcollisional arm of a hairpin in the APWP that formed as the result of orogenic collision of the juvenile THO terranes, impelled from behind by the Archean Hearne and (or) Sask craton, with the Superior craton.

© 2000 NRC Canada

Fig. 7. Apparent polar wander path (APWP) segments for the Trans-Hudson Orogen using poles for Lynn Lake gabbro pipes (LL; Dunsmore and Symons 1990); Macoun Lake pluton (MC; Symons et al. 1994); Hanson Lake pluton (HP; Gala et al. 1994); Sahli charnockite (SG; Gala et al. 1998); Reynard Lake pluton (RE; Symons 1995); Boot–Phantom pluton (BP; Symons and MacKay 1999); Molson dikes' C pole (MO_c; Zhai et al. 1994; Halls and Heaman 2000); Sudbury irruptive complex – norite and granophyre average (Morris 1984). WD is the Wapisu gneiss dome pole from this study. The wide dashed line denotes the mean 1890 to 1830 Ma trend of the APWP segments and the 1830 – 1810 Ma trend that potentially define a hairpin in the APWP from continent–continent collision in the Trans-Hudson.



Acknowledgments

The authors wish to thank David Peck and the Manitoba Energy and Mines for advice and logistical support; Kirk Stratakos for assisting in the specimen preparation; Jaros Kulhanek for helping with the magnetic measurements; and the Natural Sciences and Engineering Research Council for financial support through Lithoprobe.

References

Ague, J.J., and Brandon, M.T. 1992. Tilt and northward offset of Cordilleran batholiths resolved using igneous barometry. Nature (London), **360**: 146–149.

Ansdell, K.M., and Norman, A.R. 1993. The timing of magmatism, deformation and metamorphism along the southern flank of the Kisseynew Domain. Lithoprobe Report 34, pp. 219–225.

Ansdell, K.M., and Norman, A.R. 1995. U–Pb geochronology and tectonic development of the southern flank of the Kisseynew domain, Trans-Hudson orogen, Canada. Precambrian Research, **72**: 147–167.

Ansdell, K.M., Lucas, S.B., Connors, K., and Stern, R.A. 1995. Kisseynew metasedimentary gneiss belt, Trans-Hudson orogen (Canada): Back-arc origin and collisional inversion. Geology, **21**: 1039–1043.

As, J.A., and Zijderveld, J.D.A. 1958. Magnetic cleaning of rocks in palaeomagnetic research. Geophysical Journal of the Royal Astronomical Society, **1**: 308–319.

Ashton, K.E., Hunt, P.A., and Froese, E. 1991. Age constraints on the evolution of the Flin Flon volcanic belt and Kisseynew belt, Saskatchewan and Manitoba. *In* Radiogenic age and isotopic studies, Report 5. Geological Survey of Canada Paper 91-2, pp. 55–69.

Baldwin, D.A., Frohlinger, T.G., and Kendrick, G. 1979. Geology of the Nelson House – Pukatawagan Region (Burntwood Project). Manitoba Mineral Resources Division, Geological Maps 78-3-12 to 78-3-22 (to accompany Geological Report 78-3), scale 1 : 50 000.

Besse, J., and Courtillot, V.E. 1991. Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian plates, and true polar wander since 100 Ma. Journal of Geophysical Research, **96**: 4029–4050.

Costanzo-Alvarez, V., and Dunlop, D.J. 1988. Paleomagnetic evidence for post-2.55 Ga tectonic tilting and 1.1 Ga reactivation in the southern Kapuskasing zone, Ontario. Journal of Geophysical Research, **93**: 9126–9136.

© 2000 NRC Canada

David, J., Bailes, A.H., and Machado, N. 1996. Evolution of the Snow Lake portion of the Paleoproterozoic Flin Flon and Kisseynew belts, Trans-Hudson Orogen, Manitoba, Canada. Precambrian Research, **80**: 107–124.

Dunsmore, D.J., and Symons, D.T.A. 1990. Paleomagnetism of the Lynn Lake gabbros in the Trans-Hudson Orogen and closure of the Superior and Slave cratons. *In* The Early Proterozoic Trans-Hudson orogen of North America. *Edited by* J.F. Lewry and M.R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 215–228.

Fisher, R.A. 1953. Dispersion on a sphere. Proceedings of the Royal Society, London, Series A, **217**: 295–305.

Gala, M.G., Symons, D.T.A., and Palmer, H.C. 1994. Paleomagnetism and geotectonic implications of the Hanson Lake pluton, Hanson Lake block, Trans-Hudson orogen. Saskatchewan Geological Survey Miscellaneous Report 94-4, pp. 116–122.

Gala, M.G., Symons, D.T.A., and Palmer, H.C. 1998. Geotectonics of the Hanson Lake block, Trans-Hudson orogen, central Canada: a preliminary paleomagnetic report. Precambrian Research, **90**: 85–101.

Gordon, T.M., Hunt, P.A., Bailes, A.H., and Syme, E.C. 1990. U–Pb ages from the Flin Flon and Kisseynew belts, Manitoba: Chronology of crust formation at an Early Proterozoic accretionary margin. *In* The Early Proterozoic Trans-Hudson orogen of North America. *Edited by* J.F. Lewry and M.R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 177–199.

Graham, J.W. 1949. The stability and significance of magnetism in sedimentary rocks. Journal of Geophysical Research, **54**: 131–167.

Halls, H.C. 1998. Paleomagnetism of the Clearwater anorthosite. Lithoprobe Report 62, pp. 174–190.

Halls, H.C., and Heaman, L.M. 2000. The paleomagnetic significance of new U-Pb age data from the Molson dyke swarm, Cauchon Lake area, Manitoba. Canadian Journal of Earth Sciences, **37**: 957–966.

Harris, M.J., Symons, D.T.A., Blackburn, W.H., and Hart, C.J.R. 1997. Paleomagnetic and geobarometric study of the mid-Cretaceous Whitehorse pluton, Yukon Territory. Canadian Journal of Earth Sciences, **34**: 1379–1391.

Harris, M.J., Symons, D.T.A., Blackburn, W.H., Peck, D.C., and Turek, A. 1998. Tectonic analysis of the accretion of the Trans-Hudson orogen to the Superior craton from paleomagnetism with geobarometric corrections: Progress report (parts of NTS area 63O, 63P, 64A). *In* Manitoba Energy and Mines, Report of Activities 1998, pp. 66–68.

Hunt, P.A., and Roddick, J.C. 1988. A compilation of K–Ar ages, Report 18. *In* Radiogenic age and isotopic studies, Report 2. Geological Survey of Canada, Paper 88-2, pp. 127–153.

Hunt, P.A., and Schledewitz, D.C.P. 1992. U–Pb zircon age for a quartz porphyry in the Thunderhill Lake area, Kisseynew gneiss belt, Manitoba. *In* Radiogenic age and isotopic studies, Report 5. Geological Survey of Canada, Paper 91-2, pp. 51–54.

Irving, E., and Park, J.K. 1972. Hairpins and superintervals. Canadian Journal of Earth Sciences, **10**: 1318–1324.

Kirschvink, J.L. 1980. The least squares line and plane and the analysis of palaeomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**: 699–718.

Lucas, S.B., White, D., Hajnal, Z., Lewry, J.F., Green, A., Clowes, R., Zwanzig, H., Ashton, K., Schledewitz, D., Stauffer, M., Norman, A., Williams, P.F., and Spence, G. 1994. Three dimensional collisional structure of the Trans-Hudson orogen, Canada. Tectonophysics, **232**: 161–178.

Machado, N., and Zwanzig, H. 1995. U–Pb geochronology of the Kisseynew domain in Manitoba, provenance ages for metasediments and timing of magmatism, in Trans-Hudson orogen transect. Lithoprobe Report 48, pp. 133–138.

Manitoba Energy and Mines 1989. Bedrock Geology Compilation Map Series, Preliminary Edition, Nelson House, NTS Map 630, scale 1 : 250 000.

McElhinny, M.W. 1964. Statistical significance of the fold test in palaeomagnetism. Geophysical Journal of Royal Astronomical Society, **8**: 338–340.

Meyer, M.T., Bickford, M.E., and Lewry, J.F. 1992. The Wathaman Batholith: An Early Proterozoic continental arc in the Trans-Hudson orogenic belt. Bulletin of the Geological Society of America, **104**: 1073–1085.

Morris, W.A. 1984. Paleomagnetic constraints on the magnetic, tectonic and metamorphic evolution of the Sudbury Basin. Ontario Geological Survey, Special Vol. 1, pp. 411–427.

Stauffer, M.R. 1984. Manikewan: an early Proterozoic ocean in central Canada, its igneous history and orogenic closure. Precambrian Research, **25**: 252–281.

Symons, D.T.A. 1991. Paleomagnetism of the Proterozoic Wathaman Batholith and the suturing of the Trans-Hudson Orogen in Saskatchewan. Canadian Journal of Earth Sciences, **28**: 1931–1938.

Symons, D.T.A. 1995. Paleomagnetism of the 1851 Reynard Lake pluton, Flin Flon Domain, Trans-Hudson Orogen: Geotectonic implications. Saskatchewan Geological Survey Miscellaneous Report 95-4, pp. 137–144.

Symons, D.T.A. 1998. Precambrian plate tectonic models: shifting the paradigm for orogens such as the Trans-Hudson in Canada. Physics and Chemistry of the Earth, **23**: 753–759.

Symons, D.T.A., and Harris, M.J. 1998. Paleomagnetic data from the Big Island gneiss dome at Kisseynew Lake (NTS 63 N/3). *In* Manitoba Energy and Mines, Report of Activities 1998, pp. 25–31.

Symons, D.T.A., and MacKay, C.D. 1999. Paleomagnetism of the Boot–Phantom pluton and the amalgamation of the juvenile domains in the Paleoproterozoic Trans-Hudson orogen, Canada. Basement Tectonics, **13**: 313–331.

Symons, D.T.A., and Vandall, T.A. 1990. Paleomagnetic evidence for Proterozoic tectonism in the Kapuskasing structural zone, Ontario. Journal of Geophysical Research, **95**: 19 199 – 19 211.

Symons, D.T.A., Lohnes, C.A., and Lewchuk, M.T. 1994. Geotectonics of the Lynn Lake – LaRonge arc in the Trans-Hudson orogen from paleomagnetism of the Paleoproterozoic Macoun Lake pluton. Saskatchewan Geological Survey Miscellaneous Report 94-4, pp. 123–131.

Symons, D.T.A., Radigan, S.P., and Lewchuk, M.T. 1996. Paleomagnetism of the Davin Lake granitoids, Rottenstone Domain, Trans-Hudson Orogen (part of NTS 64 D-12 and -13). *In* Summary of investigations 1996. Saskatchewan Geological Survey, Saskatchewan Energy and Mines, Miscellaneous Report 96-4, pp. 111–118.

Van der Voo, R. 1993. Paleomagnetism of the Atlantic, Tethys and Iapetus oceans. Cambridge University Press, Great Britain.

Zhai, Y., Halls, H.C., and Bates, M. 1994. Multiple episodes of dike emplacement along the northwestern margin of the Superior Province, Manitoba. Journal of Geophysical Research, **99**: 21 717 – 21 732.

Zwanzig, H.V. 1990. Kisseynew gneiss belt in Manitoba: Stratigraphy, structure and tectonic evolution. *In* The Early Proterozoic Trans-Hudson orogen of North America. *Edited by* J.F. Lewry and M.R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 95–120.

Zwanzig, H.V. 1997. Kisseynew metasedimentary gneiss belt, Trans-Hudson orogen (Canada): Back-arc origin and collisional inversion: Comment. Geology, **25**: 90–91.

© 2000 NRC Canada